



تدريـب
PLATFORM

King Abdulaziz University
Faculty of Computer and Information Technology
Computer Science Department
Software Engineering Course
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Tadreeb Project Report
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Version 1

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Phase I: Project Description

1.1 Introduction

Cooperative training is a mandatory academic requirement in universities, bridging academia and industry to equip students with practical skills. However, students face persistent challenges such as scattered training opportunities, ineffective communication with employers, and limited clarity on training expectations. This project proposes a centralized digital platform Tadreeb to streamline training discovery, enable direct student-training provider communication. This report details the problem, objectives, and scope of the solution.

This report is structured into four main phases. Phase I defines the problem, objectives, team structure, goals, and scope of the Tadreeb platform. Phase II identifies system stakeholders, outlines data-gathering techniques, and presents detailed functional and non-functional requirements as well as use case descriptions. Phase III focuses on building the system's architecture through class diagrams and object diagrams, clarifying the relationships between key entities. Phase IV presents the dynamic system behavior through sequence diagrams, state diagrams, and activity diagrams, introduces the system architecture, and details the testing strategies applied to validate the platform's performance and usability. Additionally, the report includes appendix that document work not included and references that cite the sources used throughout the project.

1.2 Project description and objectives

Tadreeb is a centralized digital platform designed to eliminate barriers between Saudi graduates and internship opportunities. It addresses critical challenges faced by graduates, including hard access to training, mismatched skills. To accomplish this, Tadreeb:

- Establish partnerships with King Abdulaziz university.
- Establish partnerships with companies to list 100+ verified internships nationwide.
- Serving 5,000+ students needing mandatory training for degree completion.

1.3 Project team

The team consists of three students, with Ammar Alharbi assuming the role of leader. Thamer Alruqi and Nawaf Alqarni have been assigned distinct phases of the project, each taking responsibility for their respective tasks. However, the team is dedicated to collaborating and supporting each other to ensure the project meets the requirements to the highest standard.

1.4 Project Goals

- Simplify the internship search process for students by centralizing opportunities on a single platform.
- Reduce the time students spend searching for internships per week.
- Increase communication between students and companies during the application journey by integrating messaging and automated notifications.
- Promote transparency by providing detailed internship descriptions (tasks, required skills, learning outcomes).
- Enhance decision-making for training providers

1.5 Scope

The scope of our project focuses on essential deliverables that align with project goals and objectives. The specific statement of the project is a user-friendly platform that connects students and training providers. Success is measured by the number of opportunities for both students and training providers. The project is achievable since there are students who need to get co-op training, and without interfering in training or financial transactions between the two parties.

1.6.1 Included

- Advanced internship search system
- Application submission and ability to track the applications
- Internship rating system
- Dashboard for internship providers
- Student Profile

1.6.2 Excluded

- Payment or financial transactions system
- Contract management or legal agreements
- Remote training or educational content

Phase I of the Tadreeb Project established a clear foundation by defining the problem, setting specific objectives, and outlining the project scope. The team identified the need for a centralized digital platform to address the challenges students face when searching for internships. By analyzing the goals, deliverables, Phase I ensured that the project started with a clear direction and a shared understanding among all members.

Phase II: Business Requirements Specification

Phase II of the Tadreeb Project delves into the detailed design and implementation aspects that will transform the initial conceptual framework into a functional and efficient digital platform. This phase focuses on identifying and defining the system stakeholders, gathering comprehensive data to understand user requirements, and establishing both functional and non-functional requirements. By meticulously analyzing use cases and potential challenges, this phase aims to ensure that Tadreeb becomes a robust solution to bridge the gap between students seeking cooperative training and companies offering valuable internship opportunities.

2.1 System Stakeholders

1. Stockholders
 - a. **Universities**
Academic institutions require students to complete mandatory training. Critical partners for platform adoption and student onboarding.
 - b. **Training Providers**
Provide internships and rely on Tadreeb to recruit qualified students. Key contributors to the platform's success.
 - c. **Students**
Primary users of the platform who search for, apply to, and engage with internships.
2. Actors
 - a. **Students**
Register, search for internships, apply, and communicate with companies.
 - b. **Training Provider**
Post internships, review applications, and manage interactions with students.
 - c. **Tadreeb Platform Admin**
Manage technical operations, security, and platform maintenance.
3. Participants
 - a. **Training Supervisors:**
Training provider mentors guiding students during internships (do not directly use the platform).
 - b. **College Training Supervisor**
University departments promote Tadreeb to students and ensure academic alignment.

2.2 Techniques for Gathering Data

To ensure the Tadreeb Platform effectively meets the needs of students and training providers, studying current business operations is a crucial data-gathering technique. This involves analyzing how students currently search for cooperative training, how companies list internship opportunities, and the communication methods used between them. By assessing the strengths and weaknesses of existing platforms, such as university career portals or independent job boards, we can identify inefficiencies, redundancies, and areas for improvement. This study helps us understand real-world challenges, such as delays in application processing, lack of transparency in internship requirements, and limited engagement between students and companies.

In addition to studying current business processes, we conducted user interviews to gain qualitative insights from students. They shared difficulties in finding suitable training opportunities, and even when they do, they often struggle with communication. A major challenge is the delay in responses from training providers or the repeated requests for documents. We now have a clearer understanding of students' needs and expectations for the platform, helping us shape it to better serve them.

2.3 Functional Requirements

To define the functional and non-functional requirements of the Tadreeb Platform, we used a method called Customer Journey, which helped us understand the interactions between students and training providers throughout the internship process. This approach allowed us to identify key system features, user needs, and areas for improvement. The functional requirements focus on core platform capabilities, such as user registration, internship listings, application management, and communication tools, while the non-functional requirements ensure security, performance, and scalability. A detailed breakdown of the Customer Journey can be found in the appendix.

FR1: Students register and login using their information

FR1.1: Students must register using their university email

FR1.2: The system verify student accounts via email

FR1.3: Students must provide their name, age, GPA, Skills

FR1.4: The system allow students to add CV

FR2: Training providers training register and login using orgnaization information

FR2.1: Training providers must using the training provider email

FR2.2: The system verify verify training providers via email

FR2.3: Training providers must provide their training provider name & logo, contact details, and a description of the training programs we offer.

FR3: The System allow training providers to add opportunities

FR3.1: Training providers must add Internship title, description, duration, date and deadline

FR3.2: Thy system will allow training providers to specify required major, skills or GPA

FR3.3: The system will allow training providers to edit or remove the opportunity

FR3.4: The system will allow training providers to add videos or documents

FR4: The system show opportunities to students

FR4.1: The system show all opportunities in a page

FR4.2: The system allow a search choice

FR4.3: The system allow filters

FR4.4: Once the student click on an opportunity the system show full information

FR4.5: The system allow an apply button for student

FR4.6: The system add an applying to student profile

FR5: The system facilitate applying process

FR5.1: The system show all students applied to the training provider

FR5.2: Training provider can filter students

FR5.3: The System allow training provider to visit student profile

FR5.4: The System allow training provider to contact with student or reject

FR5.5: The system change status of applying in both student and training provider profile

FR5.6: The system send a message to student if he get message or be rejected

FR5.6: Training provider can accept student

FR5.7: The system closes the opportunity if the number of accepted students completed.

FR6: Rating System Management

FR6.1: The system shall allow students to give feedback on the training opportunity and submit relevant evidence.

FR6.2: The system shall ensure that student evaluations are genuine.

FR6.2.1: The system shall give the admin permission to view the submitted evidence.

FR6.2.2: The system shall display the feedback if approved by the admin; otherwise, the feedback will be deleted.

2.4 Non-functional Requirements

NFR1: The system shall be available 24h

NFR2: The system shall load at least 100,000 user accounts

NFR3: The system search results shall load within 2 seconds

NFR4: The system shall implement OAuth2 authentication for secure login

NFR5: Upload files should be in specific extension to be accepted

NFR6: Automatic database backups shall be performed every 24 hours to prevent data loss

NFR7: The user interface shall be simple and intuitive

NFR8: The platform shall be responsive on mobile, tablet, and desktop devices

NFR9: The system shall allow adding new universities and training providers dynamically without major changes

The functional and non-functional requirements outline the essential features needed to make the Tadreeb Platform effective and easy to use. By using the Customer Journey method, we were able to understand the main challenges students and training providers face and design solutions that improve their experience. The next step is to develop the system by building the platform's structure, designing the database, and creating a user-friendly interface that meets these requirements.

2.5 Use Case Description

The use case in the Tadreeb platform explains how students, companies, and admins interact with the system to complete tasks like registering, searching for internships, applying, and communicating. Each use case shows what a user wants to do, how the system responds, and what happens next. It includes who is involved, what needs to be ready before starting, the steps to complete the task, possible errors, and the final result. By organizing these use cases, we make sure the platform is easy to use and helps students find training opportunities quickly and efficiently.

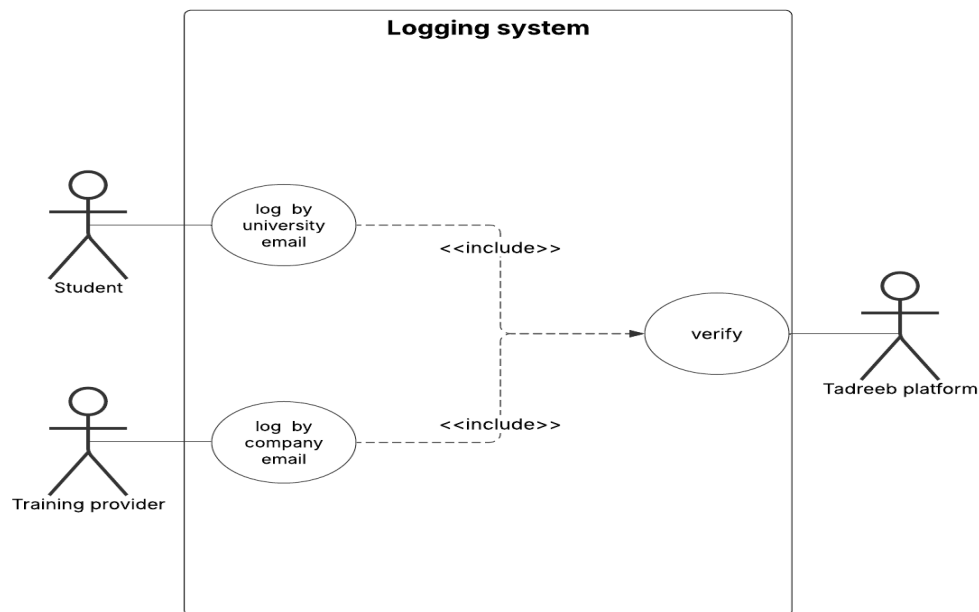


Figure 1 Use Case (1): Logging System

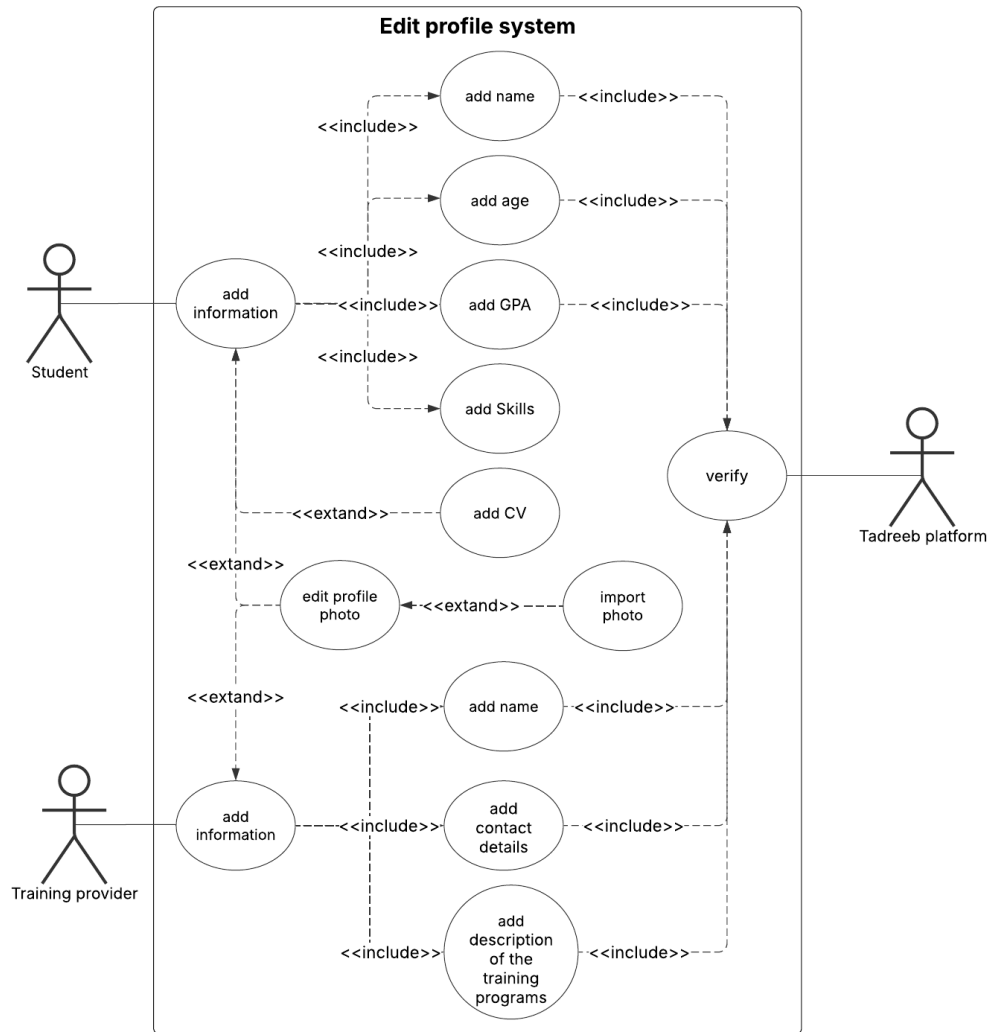


Figure 2 Use Case (II): Edit Profile System

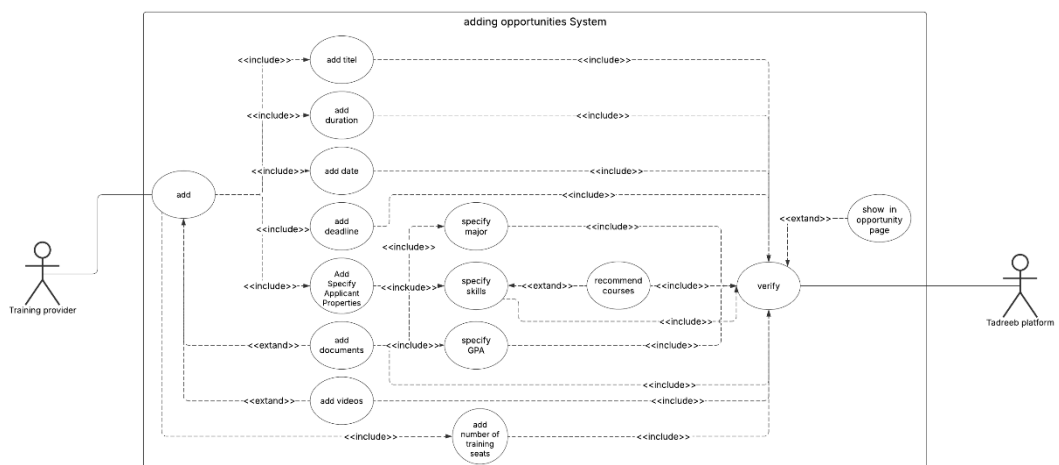


Figure 3 Use Case (III): Adding Opportunities System

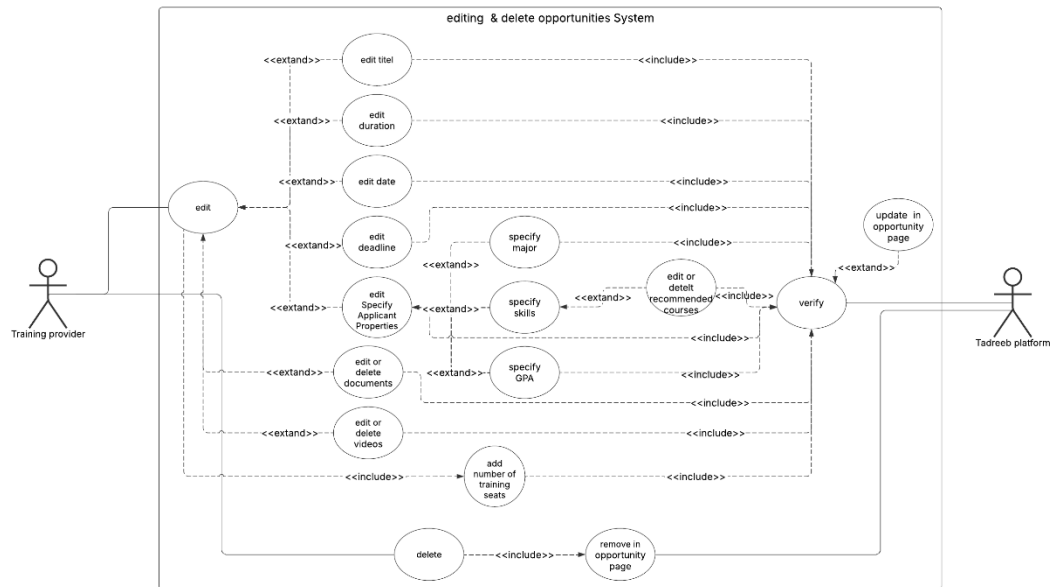


Figure 4 Use Case (IV): Editing and Deleting Opportunities System

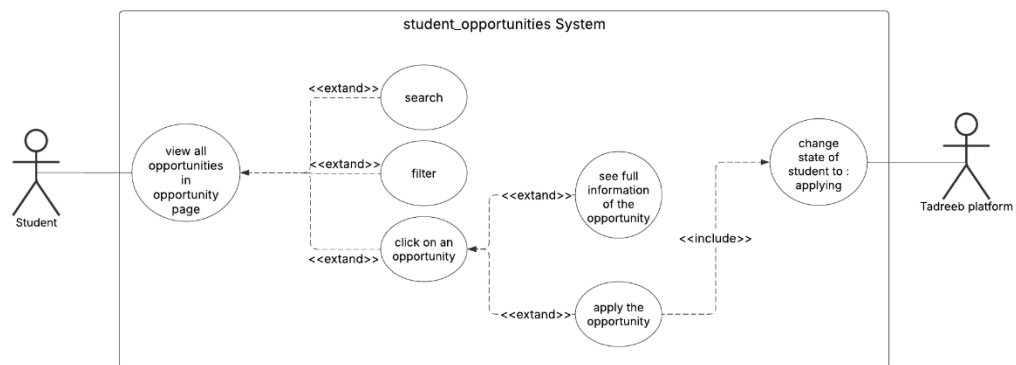


Figure 5 Use Case (V): Student Opportunities System

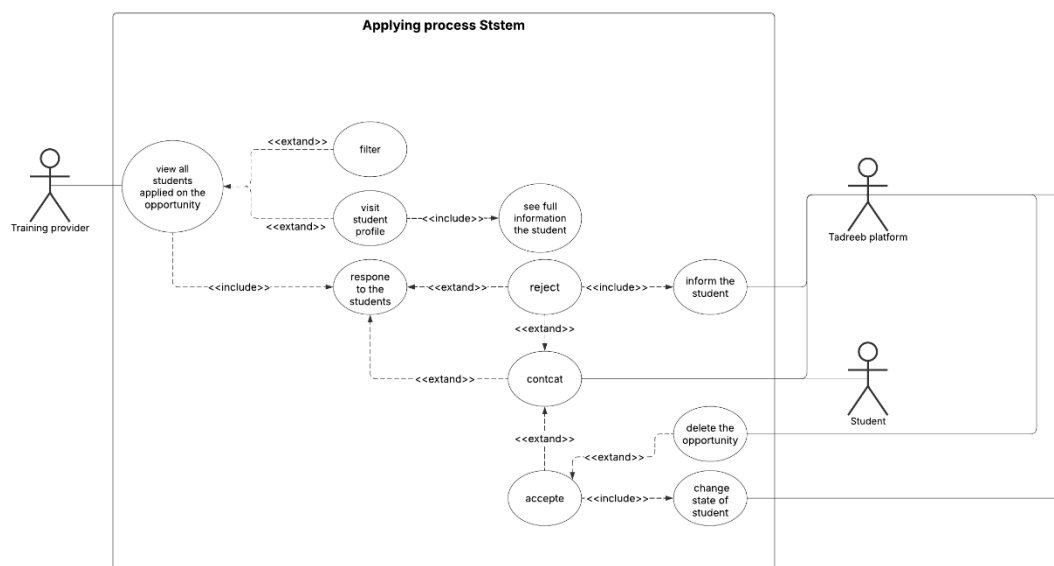


Figure 6 Use Case (VI): Applying Process System

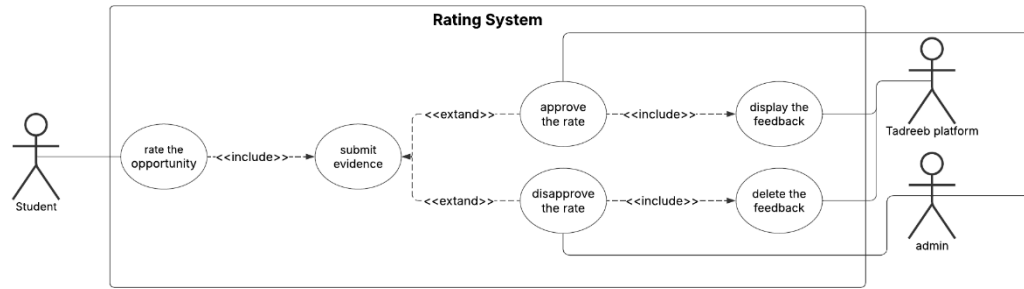


Figure 7 Use Case (VII): Rating System

After presenting the figures that visually illustrate the interaction between users and the system, we use tables to provide a detailed breakdown of each use case. The tables help organize key information, such as user actions, system responses, and possible exceptions, in a structured and clear format. We use tables because they make it easier to understand the step-by-step process of each function, ensuring that the implementation of the Tadreeb platform is well-defined and aligned with user needs. This approach allows developers to accurately translate system requirements into real features, making the platform more efficient and user-friendly.

Table 1 Use Case (I): Logging System

Logging System: UC-001	
Description: Users log in to the Tadreeb platform using their university or organization email. The system verifies the login through a hardware platform.	
Preconditions: - Users must have valid login credentials.	Postconditions: - Users are logged in and can access the platform.
Priority: High	Frequency of Use: Daily
Normal Course of Events: 1. User selects "Log in." 2. User enters a university or organization email. 3. System verifies credentials. 4. Users gain access to the platform.	Alternative Courses: - If login fails, the system prompts for re-authentication.
Exceptions: - Invalid credentials.	Includes: - Log by university email.

- System error during login.	- Log by organization email. - Verify hardware platform.
Special Requirements: - Internet connection required.	Assumptions: - Users have valid credentials.
Notes and Issues: - Ensure a secure verification process.	

Table 2 Use Case (II): Edit Profile System

Edit Profile System : UC-002	
Description: Users can edit their profiles by adding or updating personal information, skills, CV, profile photo, and contact details. Training providers can add descriptions of training programs.	
Preconditions: - User must be logged in.	Postconditions: - Profile is updated with new information
Priority: Medium	Frequency of Use: Occasionally (e.g., when profile changes are needed)
Normal Course of Events: 1. User selects "Edit Profile." 2. User updates information (name, age, GPA, skills, CV, photo, etc.). 3. User saves changes.	Alternative Courses: - If required fields are missing, the system prompts the user to fill them.
Exceptions: - System error during saving. - Invalid file format for photo or CV.	Includes: - Add name, age, GPA, skills, CV, contact details. - Edit profile photo. - Add training program descriptions (for training providers).

Special Requirements: -- Internet connection required.	Assumptions: - Users will provide valid information.
Notes and Issues: - Ensure validation for all fields.	

Table 3 Use Case (III): Adding Opportunities System

Adding Opportunities System: UC-003	
Description: Training providers can add new training opportunities by specifying details such as title, duration, date, applicant properties, deadline, and recommended courses. They can also add documents and videos.	
Preconditions: - Training provider must be logged in. - Training provider must have a verified account.	Postconditions: - Opportunity is reviewed by admin and posted on the platform.
Priority: High	Frequency of Use: Regularly (e.g., whenever new opportunities are available)
Normal Course of Events: 1. Training provider selects "Add Opportunity." 2. Provider fills in details (title, duration, date, applicant properties, etc.). 3. Provider submits the opportunity for admin review.	Alternative Courses: - If required fields are missing, the system prompts the provider to fill them.
Exceptions: - System error during submission. - No internet connection.	Includes: - Add title, duration, date, deadline. - Specify applicant properties.

	- Recommend courses. - Add documents and videos. - Verify opportunity.
Special Requirements: - Internet connection required.	Assumptions: - Admins will review opportunities promptly.
Notes and Issues: - Ensure validation for all fields.	

Table 4 Use Case (IV): Editing and Deleting Opportunities System

Editing and Deleting Opportunities System : UC-004	
Description: Training providers can edit or delete existing training opportunities. This includes modifying details such as title, duration, date, applicant properties, and recommended courses. They can also remove documents and videos.	
Preconditions: - Training providers must be logged in. - Opportunity must exist.	Postconditions: - Opportunity is updated or removed from the platform.
Priority: Medium	Frequency of Use: Occasionally (e.g., when changes to opportunities are needed)
Normal Course of Events: 1. Training provider selects "Edit Opportunity." 2. Provider modifies details (title, duration, date, etc.). 3. Provider saves changes or deletes the opportunity.	Alternative Courses: -If required fields are missing, the system prompts the provider to fill them.

Exceptions: - System error during editing or deletion. - Invalid file format for photo or CV.	Includes: - Edit title, duration, date, deadline. - Modify applicant properties. - Edit or delete recommended courses, documents, videos. - Update or remove opportunity.
Special Requirements: -- Internet connection required.	Assumptions: - Admins will review changes promptly
Notes and Issues: - Ensure validation for all fields.	

Table 5 Use Case (V): Student Opportunities System

Student Opportunities System : UC-005	
Description: Students can search, filter, and view all available training opportunities. They can click on an opportunity to see full details and apply for it. The system also changes the state of the student's application.	
Preconditions: -- Student must be logged in.	Postconditions: - Student view opportunities and apply for them.
Priority: High	Regularly (e.g., daily by students)
Normal Course of Events: 1. Student searches and filters opportunities. 2. Student clicks on an opportunity to view details. 3. Student applies for the opportunity. 4. System changes the state of the application.	Alternative Courses: - If no opportunities match the search criteria, the system displays a message.

Exceptions: - System error during search or application. - No internet connection.	Includes: - Search and filter opportunities. - View full details. - Apply for opportunity. - Change application state.
Special Requirements: - Internet connection required.	Assumptions: - Students will use the platform regularly..
Notes and Issues: - Ensure accurate search and filter functionality.	

Table 6 Use Case (VI): Applying Process System

Applying Process System : UC-006	
Description: Training providers can view all students who have applied for a specific opportunity. They can filter applications, visit student profiles, and select or reject students based on their information. The system also allows training providers to delete opportunities or change the state of a student's application.	
Preconditions: - Training provider must be logged in. - Students must have applied for the opportunity.	Postconditions: - Applications are reviewed, and students are accepted or rejected.
Priority: High	Frequency of Use: Regularly (e.g., whenever applications are submitted)

Normal Course of Events: 1. Training provider views applications. 2. Provider filters applications and visits student profiles. 3. Provider accepts or rejects students.	Alternative Courses: - If no students meet the criteria, the system displays a message.
Exceptions: - System error during application review. - No internet connection.	Includes: - View and filter student applications. - Visit student profiles. - Accept or reject students. - Delete opportunities or change application state.
Special Requirements: -- Internet connection required.	Assumptions: - Training providers will review applications promptly..
Notes and Issues: -Ensure accurate filtering and application review functionality	

Table 7 Use Case (VII): Rating System

Rating System : UC-007	
Description: Students can rate training opportunities and submit evidence to support their ratings. Admins can approve or disapprove the ratings, and feedback is displayed on the platform. Admins also have the ability to delete feedback.	
Preconditions: - Student must be logged in.	Postconditions: - Ratings and feedback are displayed on the platform.

- Student must have participated in the opportunity.	
Priority: Medium	Frequency of Use: Occasionally (e.g., after completing an opportunity)
Normal Course of Events: 1. Student rates the opportunity and submits evidence. 2. Admin reviews and approves or disapproves the rating. 3. Feedback is displayed on the platform.	Alternative Courses: - If the rating is disapproved, the system notifies the student.
Exceptions: - System error during rating submission. - No internet connection.	Includes: - Rate opportunity and submit evidence. - Approve or disapprove ratings (admin). - Display and delete feedback (admin).
Special Requirements: -- Internet connection required.	Assumptions: - Admins will review ratings promptly.
Notes and Issues: - Ensure accurate rating and feedback functionality.	

2.6 Sources of Domain Analysis Information

The domain analysis data for our system is gathered from multiple sources to ensure that Tadreeb effectively addresses the challenges faced by students and training providers in securing internships.

- **Government Regulations & Policies:** The Ministry of Education (MOE) mandates that university students must complete cooperative training as a graduation requirement. Additionally, the Ministry of Human Resources and Social Development (MOHRSD) requires medium and large companies to provide internship opportunities for students ⁽²⁾. These regulations form the foundation of our platform.
- **University Internship Requirements:** Data from the **Council of University Affairs** indicates that approximately quarter of a million students are currently enrolled in Makkah Region universities ⁽³⁾, all of whom will require internships during their studies. University career centers and academic advisors provide additional insights into the challenges students face when securing internships.
- **Industry & Employer Insights:** To understand the perspective of training providers, we analyzed internship postings from local companies, corporate training programs, and recruitment platforms. Interviews with HR professionals and internship coordinators highlighted the difficulties employers face in managing and selecting student applicants.
- **Student Feedback & Surveys:** Direct input from students at universities was collected to identify pain points in the internship search process. Common challenges include lack of centralized opportunities, poor communication with companies, and unclear training expectations. This data played a crucial role in shaping Tadreeb's features.

Phase II has successfully built upon the groundwork established in Phase I by diving deep into the technical and functional aspects of the Tadreeb platform. The comprehensive analysis of system stakeholders, data gathering techniques, and detailed functional and non-functional requirements has paved the way for a well-structured and user-centric platform. By addressing the real-world challenges faced by students and training providers, Phase II has ensured that Tadreeb will be a valuable tool for bridging the gap between academia and industry. The detailed use case descriptions and domain analysis have provided a solid blueprint for the platform's development, ensuring that it will meet the needs of its users effectively. As we move forward, the insights gained in Phase II will guide the development and implementation process, bringing us closer to achieving the project's goals

Phase III: Design and Structuring

In the third phase of the project, the focus was specifically on designing the UML Class Diagram. This diagram acts as a cornerstone in the design phase, outlining the system's general structure and emphasizing the relationships between different objects. The primary goal of this design is to ensure organized data handling and clarify interactions between users and the platform's features, setting the stage for a robust and efficient system

3.1 Class Diagram

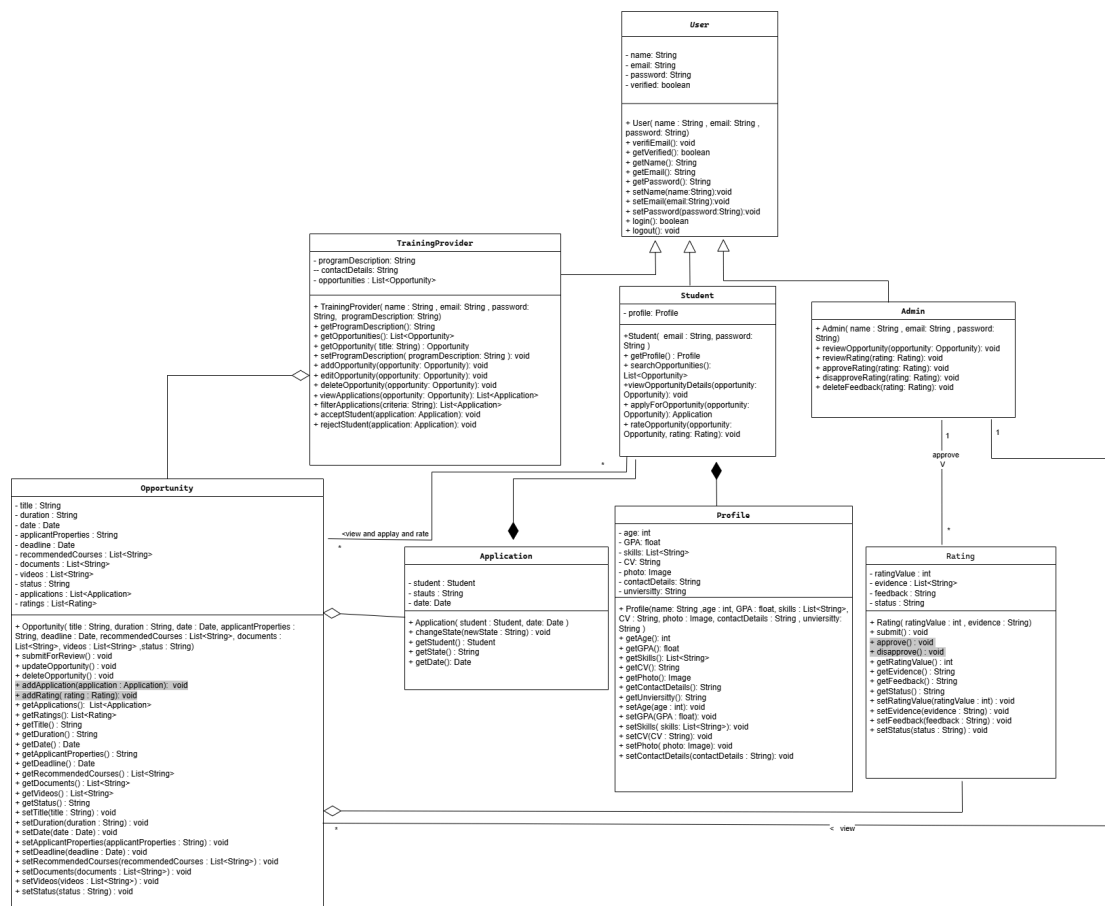


Figure 8 Project Class Diagram

3.2 Class Association and Multiplicity

Associations

- TrainingProvider – Opportunity
 - A TrainingProvider is responsible for posting internship opportunities.
- Student – Opportunity
 - A Student interacts with the platform by browsing, selecting, and applying for internship opportunities.
- Student – Application
 - a student submits an application to an opportunity.
- Opportunity – Application
 - An opportunity can receive many applications from students. Each application is tied to one specific opportunity.
- Student – Rating
 - A student can submit a rating regarding the opportunity.
- Opportunity – Rating
 - An opportunity may accumulate multiple ratings from students.

Multiplicity

- TrainingProvider and Opportunity:
 - A training provider can post zero or more opportunities
 - Each opportunity must be associated with exactly one training provider
- Student and Opportunity:
 - A student may apply for or interact with zero, one, or many opportunities.
 - An opportunity might be applied to by zero, one, or many students.
- Student and Rating:
 - A student might write none or several ratings based on their training experiences.
 - Each rating is authored by exactly one student.
- Opportunity and Rating:
 - An opportunity may have received no ratings or multiple ratings from different students.
 - Each individual rating corresponds to one specific opportunity.

3.3 Object Diagram

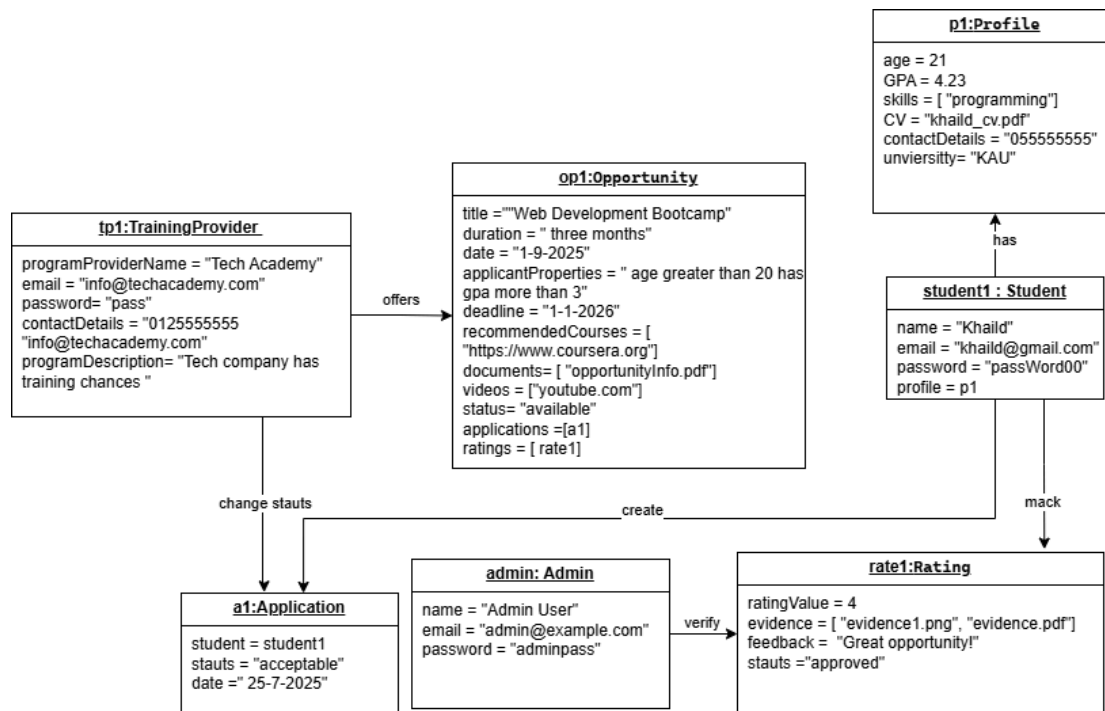


Figure 9 Project Object Diagram

With the completion of the third phase and the development of the UML Class Diagram, a comprehensive vision of the system's architecture and core element relationships has been established. This achievement marks a critical step toward translating the project's vision into a real-world software solution. The diagram provides the foundation for constructing the remaining parts of the system in a way that effectively supports the project's objectives and meets user needs.

Phase IV: Modeling, Interaction and Behavior Diagram and Testing.

In phase IV of our project, we focused on transforming our design concepts into interactive and behavior-based models to represent how the system operates in real-world scenarios. This includes creating the sequence diagram, state diagram, and activity diagram. These models help us understand how different components of Tadreeb communicate and respond to events during user sessions. We also explored the overall system architecture to illustrate how various modules are connected. Finally, we conducted initial testing to ensure that the platform behaves as expected under different conditions.

4.1 Sequence Diagram

Sequence diagrams are important for visualizing how different parts of the system interact over time. They help in understanding the flow of messages between users and the platform, clarifying the processes that take place during key operations such as registration, applying for opportunities, and managing applications. By using sequence diagrams, we can ensure that each interaction is logically organized and supports the platform's functional requirements.

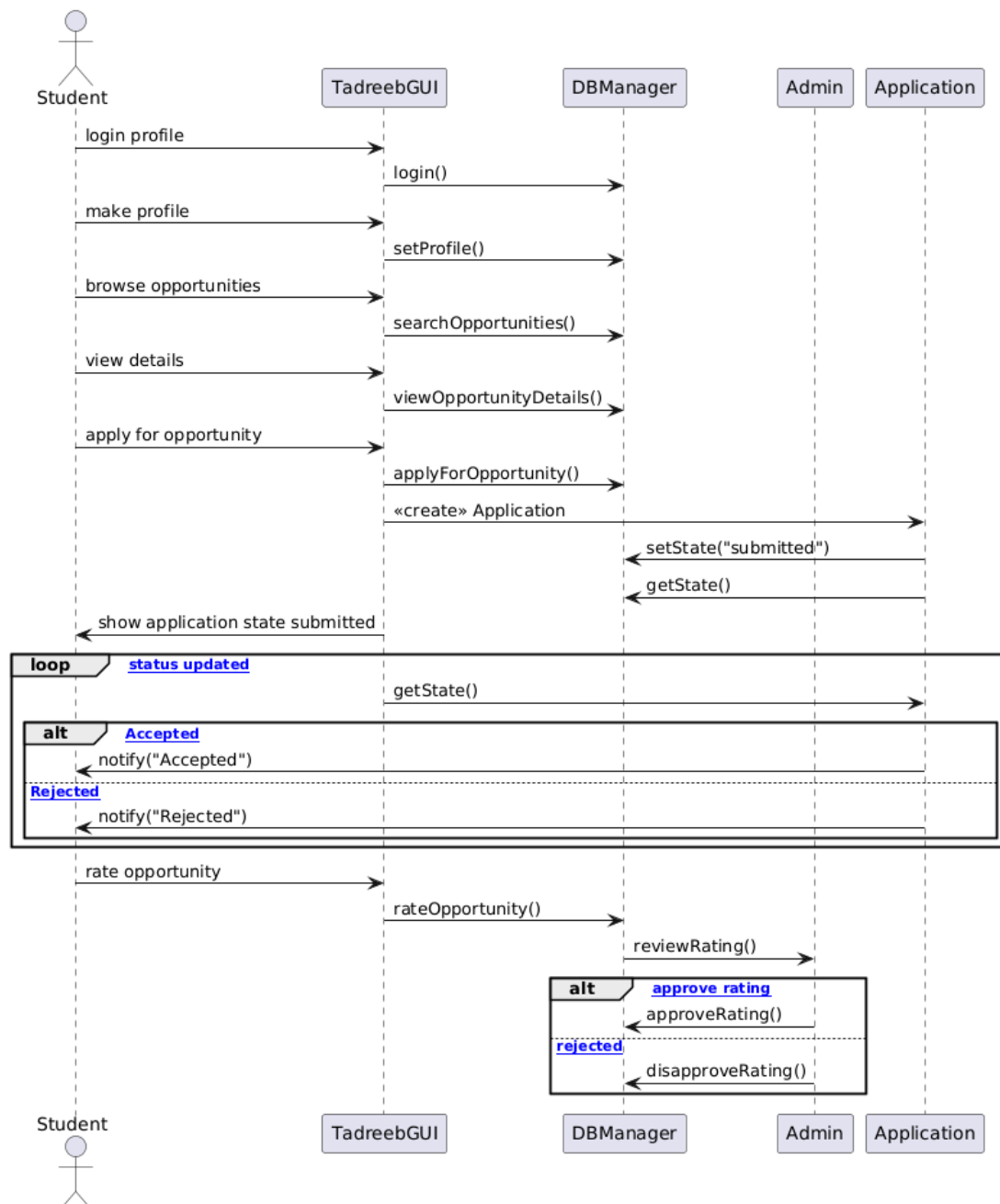


Figure 10 Sequence Diagram for Student Interaction

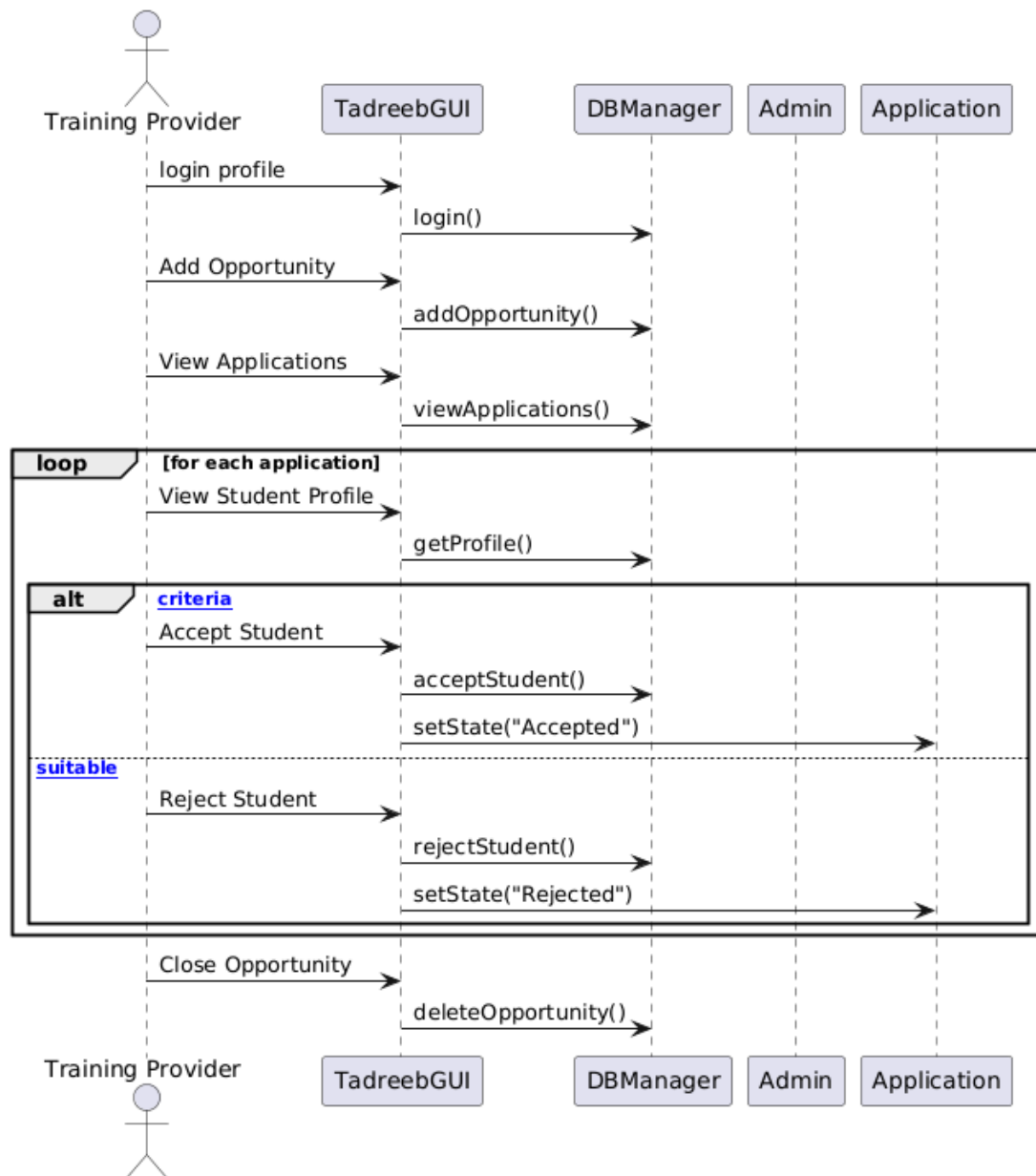


Figure 11 Sequence Diagram for Training Provider Interaction

The sequence diagrams developed illustrate the key interactions between users and the Tadreeb platform. *Figure 10* presents the sequence of actions a student performs when logging in, browsing opportunities, applying for training, and receiving status updates. *Figure 11* shows the steps a training provider follows when posting opportunities, reviewing applications, and managing communication with students. Together, these diagrams provide a clear view of how the system supports both students and training providers during the internship process.

4.2 State Diagram

State diagrams are important for understanding how different system components behave based on user actions and internal events. They show the various states an object can be in and the transitions between these states. Using state diagrams helps ensure that the platform responds correctly to different situations, such as application submission, opportunity management, and rating approval.

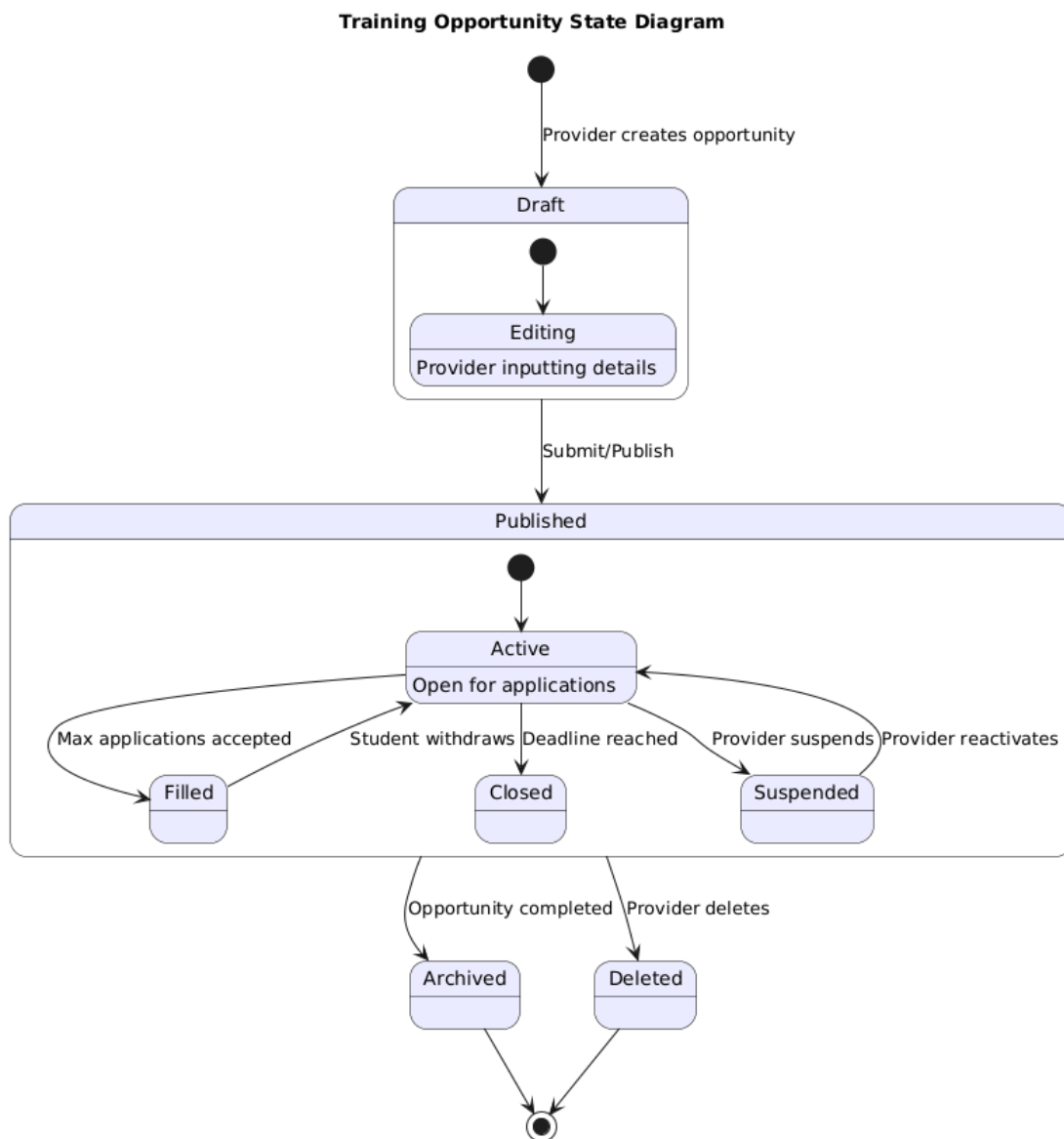


Figure 12 State Diagram for Training Opportunity Lifecycle

Student Application State Diagram

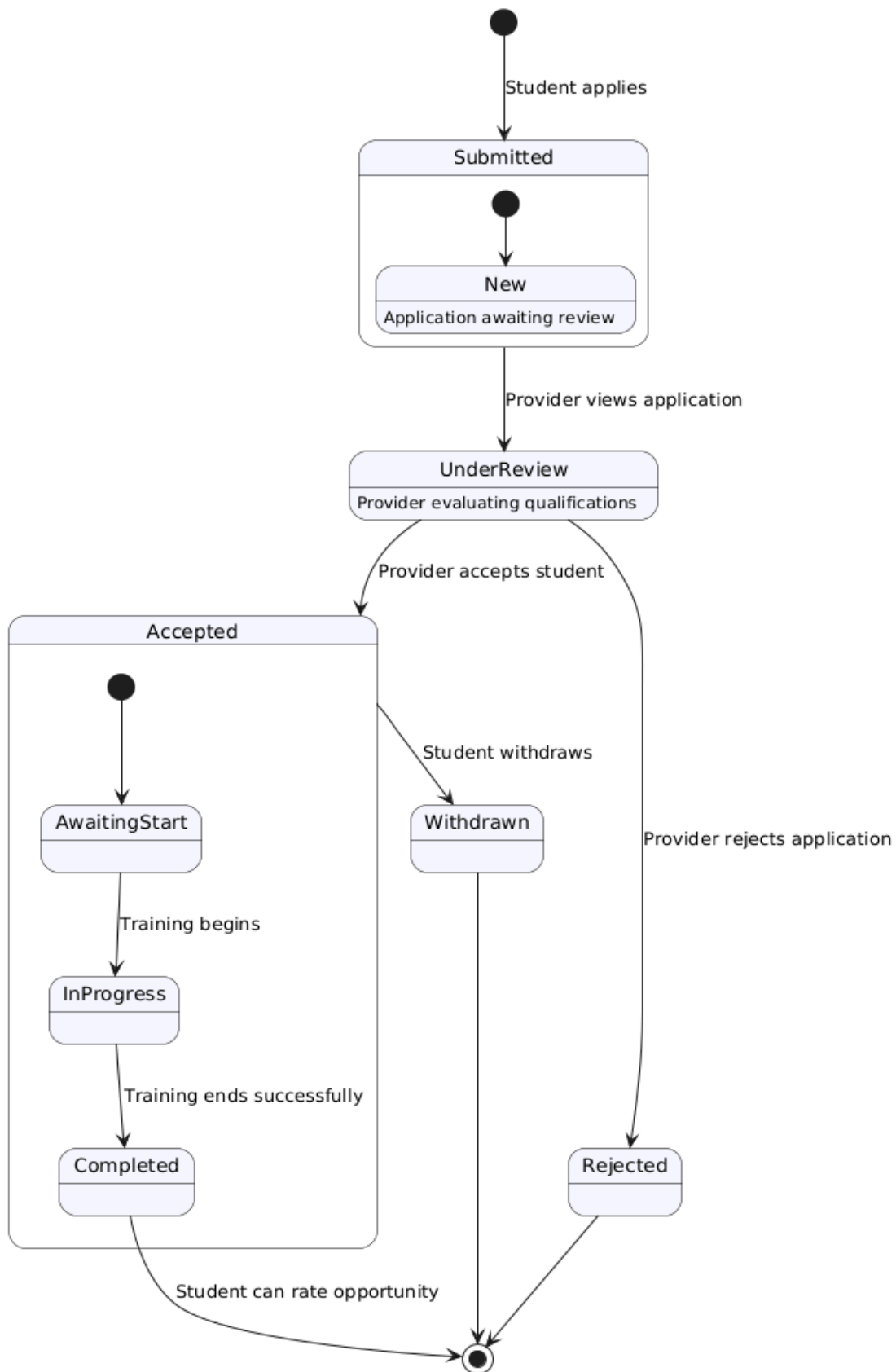


Figure 13 State Diagram for Student Application Lifecycle

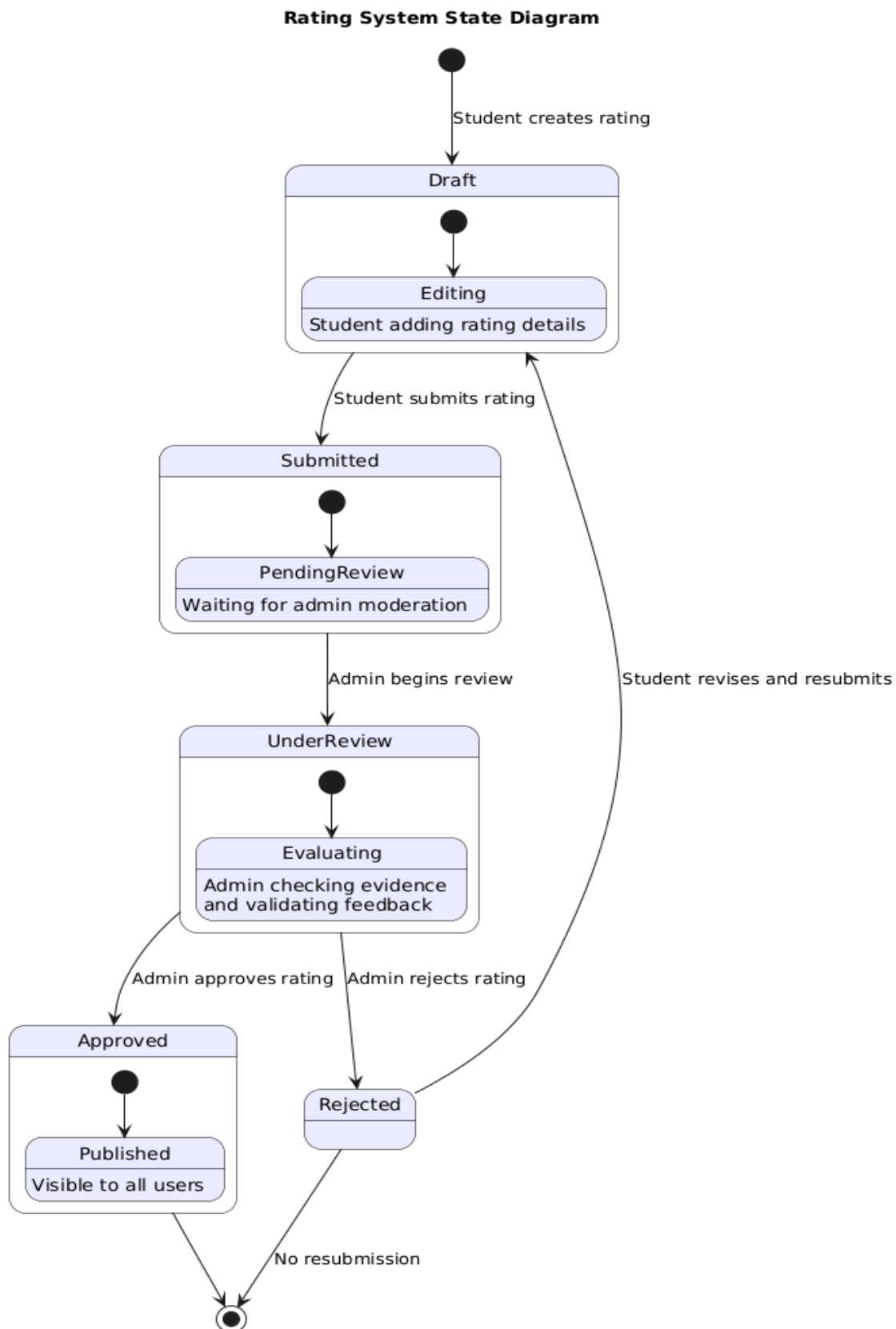


Figure 14 State Diagram for Rating Submission Process

The state diagrams illustrate the behavior of key entities within the Tadreeb platform. *Figure 12* shows the different states of a training opportunity, including its creation, activation, and closure. *Figure 13* presents the states an application goes through, from submission to acceptance or rejection. *Figure 14* displays the process of submitting, reviewing, and publishing a rating. These diagrams help visualize the complete life cycle of main processes and ensure that all transitions between states are handled properly.

4.3 Activity Diagram

Activity diagrams are useful for modeling the workflow of different processes in the system. They show the sequence of actions and decisions that take place during important operations. By using activity diagrams, we can better understand how users move through tasks such as registering, creating opportunities, applying for training, and providing ratings, ensuring that each step is clear and logically connected.

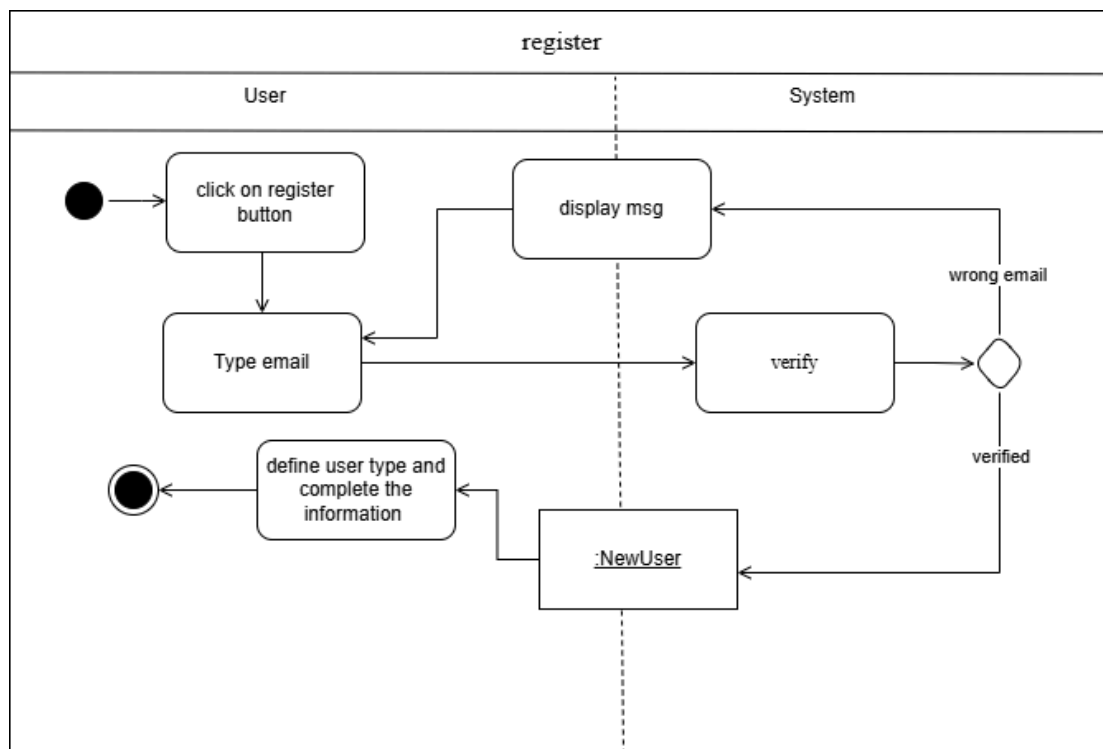


Figure 15 Activity Diagram for Student Registration Process

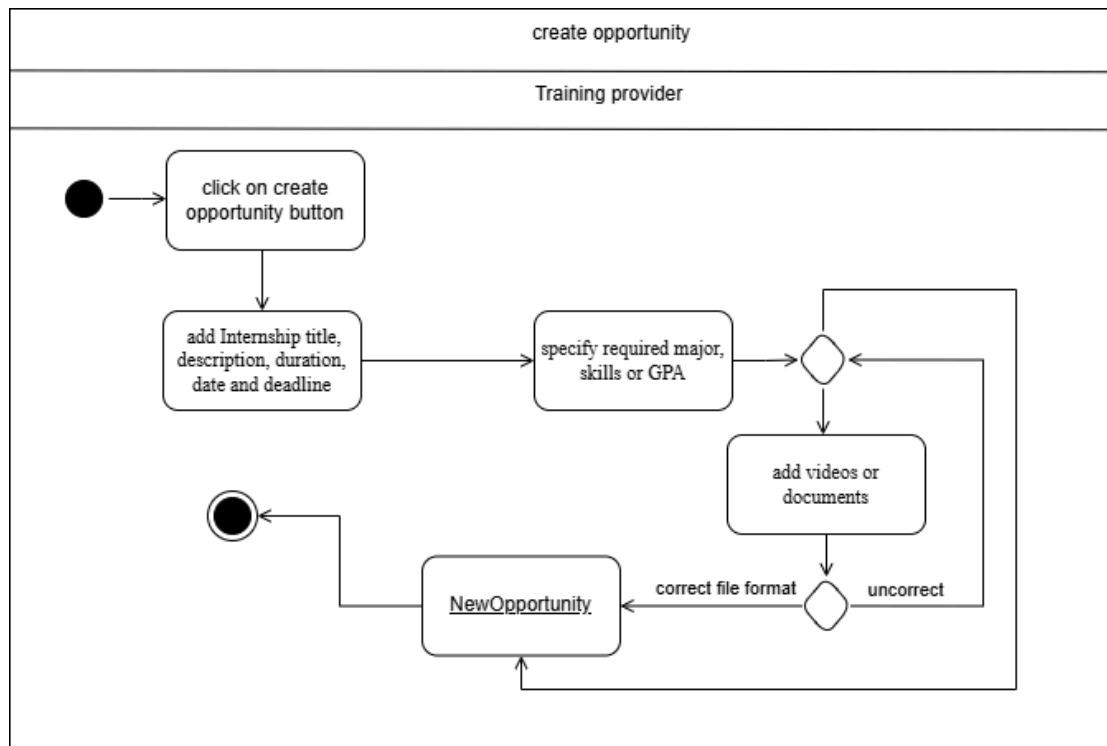


Figure 16 Activity Diagram for Creating a Training Opportunity

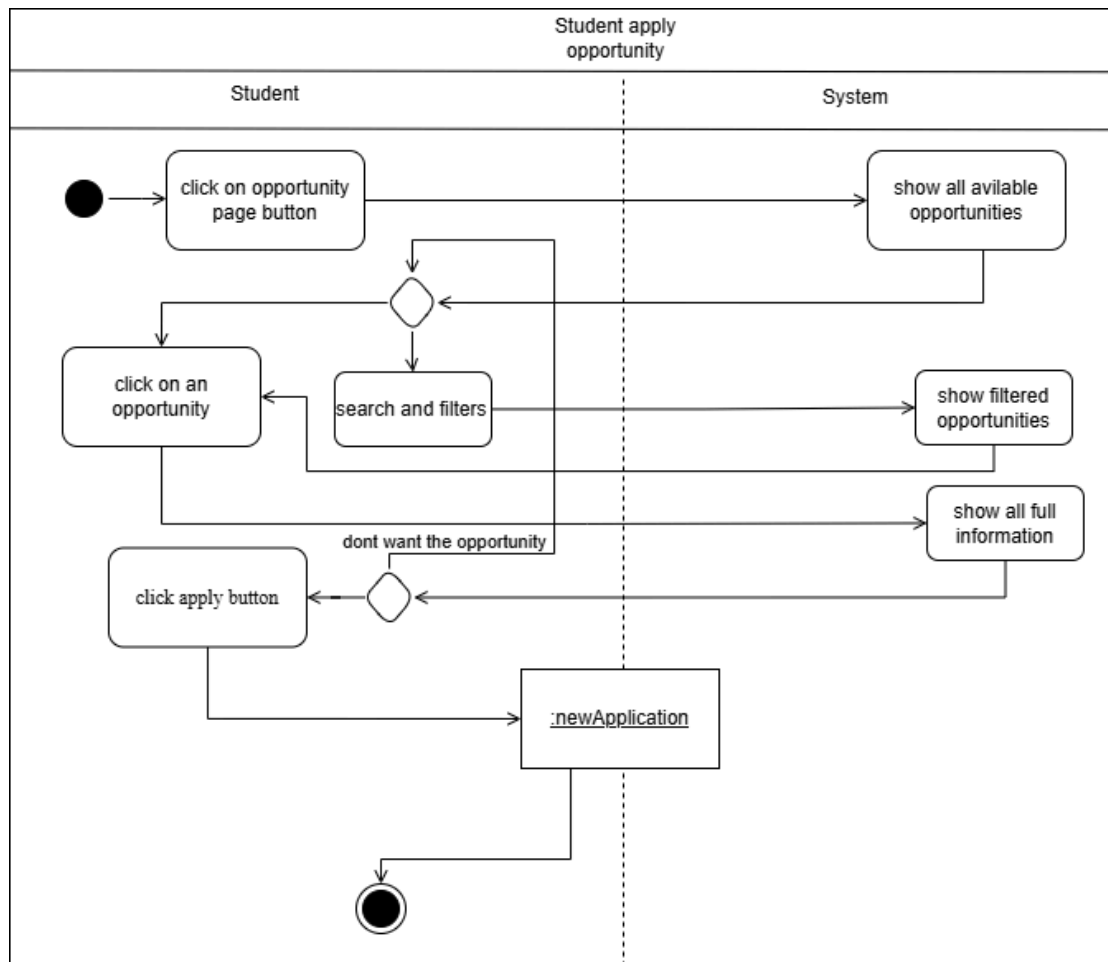


Figure 17 Activity Diagram for Student Applying for an Opportunity

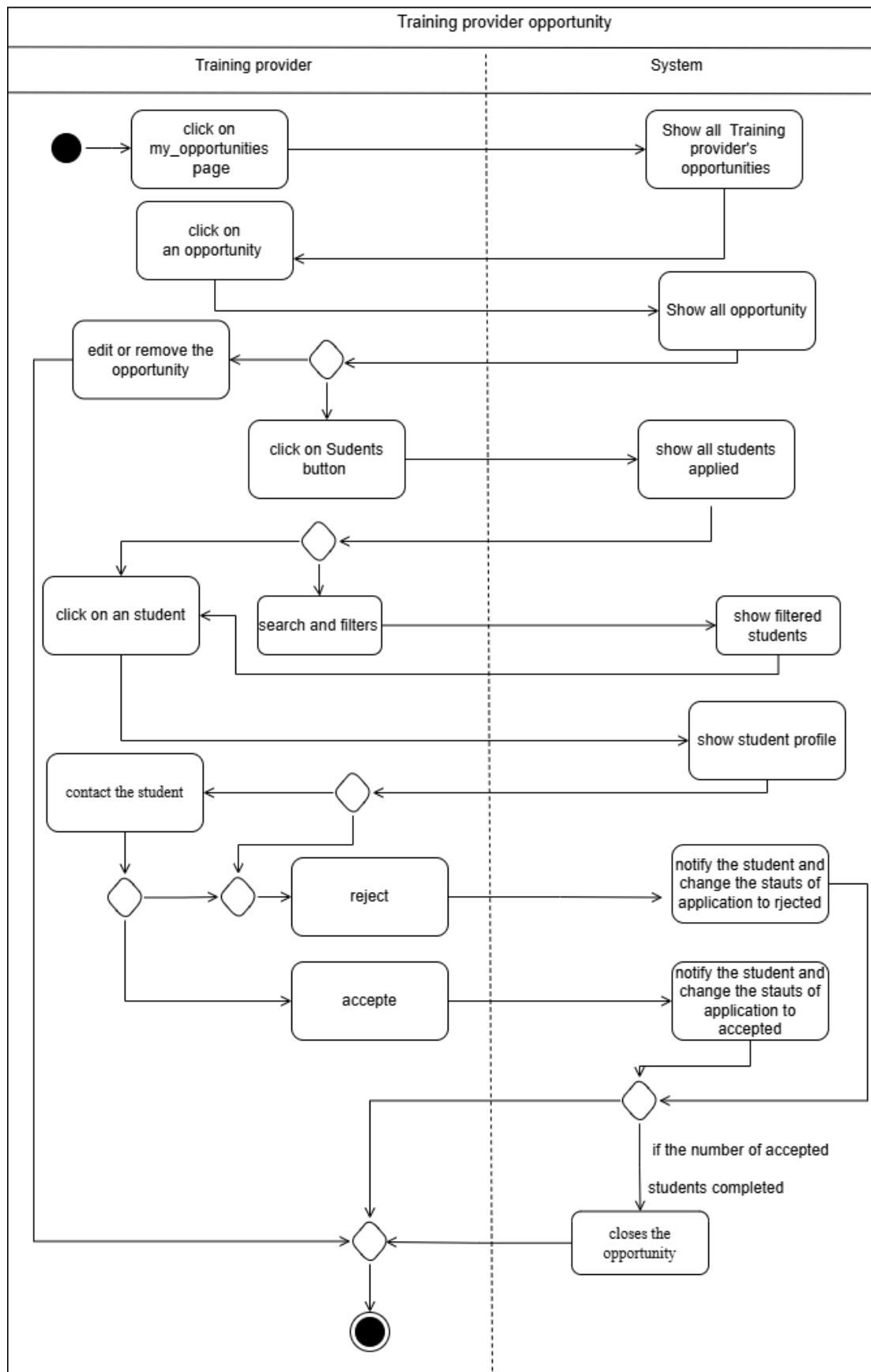


Figure 18 Activity Diagram for Training Provider Managing Opportunities

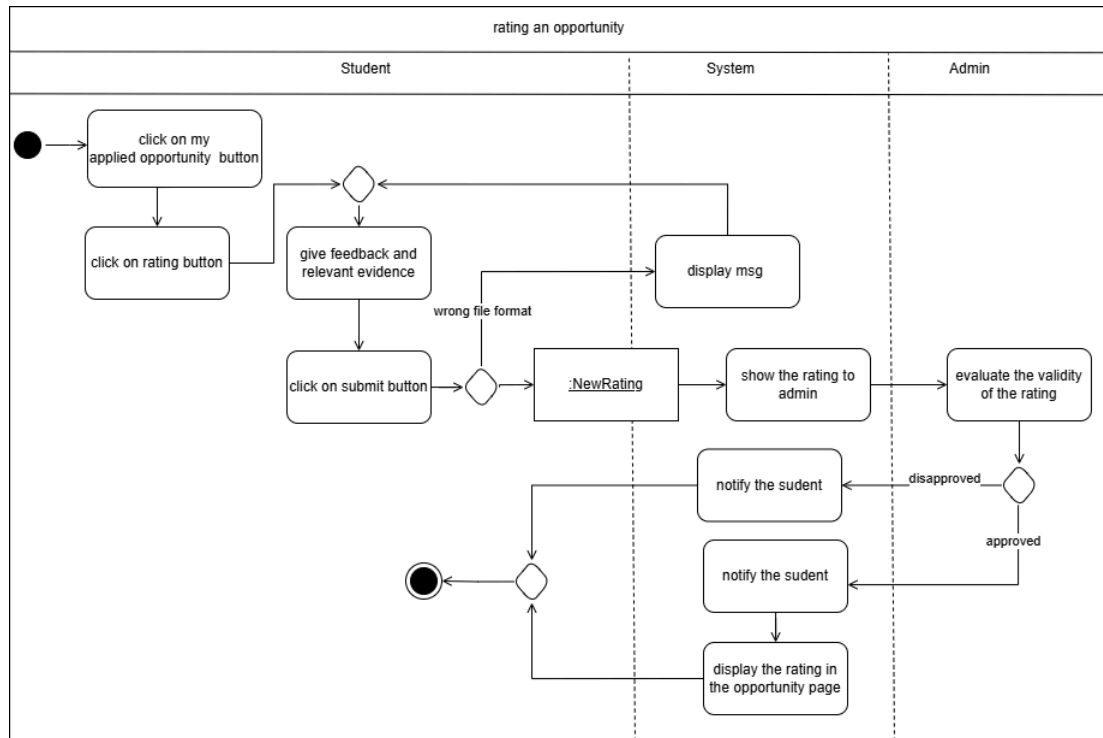


Figure 19 Activity Diagram for Student Rating Submission

The activity diagrams illustrate the workflows of major processes in the Tadreeb platform. *Figure 15* shows the steps involved in the student registration process. *Figure 16* details the process of a training provider creating a new opportunity. *Figure 17* explains how a student applies for an opportunity. *Figure 18* shows how a training provider manages posted opportunities. *Figure 19* outlines the steps students follow when submitting ratings. Together, these diagrams provide a full overview of the activities carried out by users and ensure the system supports each operation in an organized way.

4.4 System Architecture

The system architecture diagram shows how the main parts of Tadreeb are connected. It explains the flow of data between users, the platform, and the database. Clear architecture helps in building a secure and scalable system.

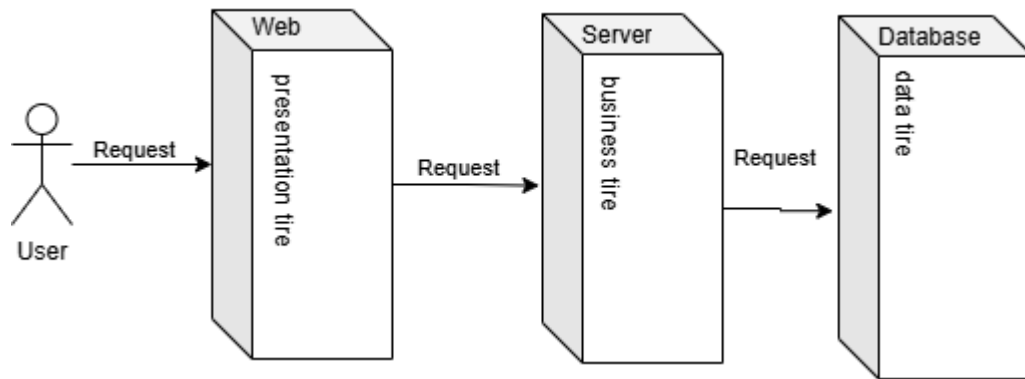


Figure 20 System Architecture Diagram for Tadreeb Platform

Testing

Objectives

The goal of testing Tadreeb is to make sure all key processes work smoothly and efficiently for both students and training providers. This includes:

- Making sure users can register without issues.
- Allowing training providers to create internship opportunities easily.
- Ensuring students can apply for opportunities correctly.
- Helping training providers manage applications properly.
- Allowing students to rate their internship experiences effectively.

Strategy

The purpose of this testing strategy is to ensure that the Tadreeb platform operates efficiently, meets all predefined requirements, and addresses potential issues before deployment, providing a seamless, secure, and user-friendly experience for students and training providers. To achieve this, several core functions are tested to verify accuracy and reliability. These include user registration, ensuring students and training providers can sign up correctly; opportunity creation, validating that training providers can list internship opportunities with complete and accurate details; student applications, confirming that students can search, apply, and track internship opportunities effortlessly; training provider management, ensuring providers can review and respond to student applications effectively; and the internship rating system, allowing students to provide meaningful feedback with proper validation. To assess system behavior and performance, the black-box testing method is used, focusing on how the platform responds to user inputs without analyzing its internal code. This approach ensures that Tadreeb functions as expected while prioritizing usability, security, and efficiency.

Below are the key tests for each process:

1. Registration Process:

Test	Expected Result
Register user with valid email	Account successfully created
Register with invalid email	Error message shown
Register without mandatory details	Validation error shown
System sends verification email	Email received

Table 8 Test Cases for Registration Process

2. Create Opportunity

Test	Result Expected
Provider adds new opportunity	Opportunity listed successfully
Provider submits incomplete details	Validation error displayed
Provider uploads files	Files uploaded successfully
Provider specifies skills & GPA	Students can filter opportunities correctly

Table 9 Test Cases for Creating Opportunities

3. Student Apply for Opportunity

Test	Expected Result
Student applies for an internship	Application recorded and displayed in profile
Student applies without required info	System prevents submission
Application status updates when accepted/rejected	Status changes for both student and provider

Table 10 Test Cases for Student Applying for Opportunities

4. Training Provider Manages Applications

Test	Expected Result
Provider views student applications	List of applicants displayed correctly
Provider filters applicants	Only matching students shown
Provider accepts/rejects an application	Student receives a notification

Table 11 Test Cases for Training Provider Managing Applications

5. Rating an Internship

Test	Expected Result
Student submits valid rating	Rating recorded successfully
Student submits without evidence	System asks for additional documentation
Admin approves rating	Rating displayed publicly
Admin rejects invalid rating	Rating removed

Table 12 Test Cases for Rating an Internship

A well-defined testing strategy is crucial to ensuring Tadreeb functions efficiently, meets user needs, and maintains system reliability. By testing core processes such as registration, opportunity creation, student applications, training provider management, and internship rating, we validate the platform's ability to deliver a smooth and effective experience. Implementing black-box testing further ensures accurate system responses while maintaining usability and security. With this approach, Tadreeb can confidently support students and training providers, enhancing the internship search and management process.

Conclusion

The Tadreeb project aimed to solve the challenges students face in finding cooperative training by building a centralized platform. In Phase I, the project objectives, scope, and team structure were defined clearly to align with academic and industry needs. Identifying problems such as scattered opportunities and poor communication established a strong foundation for focused system development.

In Phase II and Phase III, the business requirements were gathered using real user feedback, and system requirements were detailed through functional and non-functional specifications. Use cases were developed to guide system behavior, and the class and object diagrams were designed to organize platform entities such as students, training providers, and opportunities. The focus during these phases was to ensure a clear, scalable system structure that meets user needs.

Phase IV concentrated on modeling system interactions through sequence diagrams, state diagrams, and activity diagrams. Testing was carried out using black-box methods to verify key processes like registration, opportunity creation, application, and feedback submission. The project concluded with a platform that addresses real-world cooperative training challenges, delivering a structured and user-centered solution.

Appendix

Customer Journey – Student:

I am Khaled, a student at the University of King Abdulaziz. I need to secure a training position for this summer, which is in three months. I recently discovered the Tadreeb platform. Here's how my experience went:

1. I opened the Tadreeb platform, chose my university, and then logged in using my university login system.
2. After logging in, I clicked on the profile button and found all my information: name, age, GPA. There was also space to upload my CV, certificates, and list my skills.
3. Next, I clicked on the opportunities button and found a page with numerous companies offering training seats.
4. I clicked on the XXX company and saw the requirements they had, as well as the benefits they provided and suggested courses to complete the requirements.
5. Luckily, I had enough time to take the suggested course. After completing it, I applied for a seat at XXX company.
6. After a short period, they accepted my application, and we communicated through the platform for the interview, which I successfully passed.

Once my training journey concluded, I returned to the platform to rate the XXX company. There were multiple aspects to rate, such as salary, work environment, tasks, benefits, and more. They asked me to provide evidence for each significant rating. A week later, I saw my ratings for the company on the platform.

Customer Journey – Training Provider:

As a training provider on the Tadreeb platform, our primary goal is to find and nurture talented students from various universities. Here's my journey in the platform:

1. Registration and Profile Setup:
 - a. I registered my company on the Tadreeb platform by providing the necessary information, including company name, contact details, and a description of the training programs we offer.
 - b. Next, I set up the company profile, uploading our logo, mission statement, and detailed descriptions of the available training opportunities.
2. Creating Training Opportunities:
 - a. I listed the training positions available at our company, specifying the requirements, duration, benefits, and any courses that would help applicants meet the qualifications.
 - b. I also uploaded supporting documents and videos that provide more insight into the company culture and expectations.

3. Reviewing Applications and Conducting Interviews:
 - a. Once students started applying, I accessed their profiles to review their CVs, certificates, and skills.
 - b. I shortlisted candidates who met our criteria and sent them messages through the platform to schedule interviews.
 - c. After conducting the interviews, I evaluated the performance of each candidate and selected the best fit for our training program.
4. Onboarding and Training:
 - a. I provided the selected students with onboarding materials and schedules through the platform.
 - b. Throughout the training period, I used the platform to monitor their progress, provide feedback, and offer additional resources to ensure a successful training experience.
5. Receiving Feedback
 - a. After the training concluded, I encouraged the students to rate their experience with our company on the platform.
 - b. I reviewed their ratings and feedback to understand their perspective and identify areas for improvement.

References

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2. <https://www.argaam.com/ar/article/articledetail/id/1696144>
3. <https://www.cua.gov.sa> statics until 2025 January